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BlockChain & DLT

Experiment No: 7

AIM:

To implement MetaMask wallet integration and perform transactions using Solidity for **Academic Credentials Verification System**

.

Task to be Performed:

- Install MetaMask
- Create or Import Account
- Connect to Sepolia Testnet
- Fund wallet using Faucet
- Connect MetaMask with Remix IDE
- Create Smart Contract for Academic Credentials
- Compile and Deploy Smart Contract
- Perform Transactions
- Verify stored data

Theory:

MetaMask:

MetaMask is a cryptocurrency wallet that allows users to interact with blockchain networks. It acts as a bridge between the browser and blockchain, enabling transactions and smart contract interaction.

Testnet:

A testnet is a blockchain network used for testing purposes. It uses fake cryptocurrency like Sepolia ETH, allowing developers to test smart contracts without real money.

Academic Credential Verification using Blockchain:

In this system, student academic details such as name, course, and grade are stored on the blockchain using a smart contract. This ensures data is secure, tamper-proof, and easily verifiable.

Advantages:

- Secure and tamper-proof records
- Easy verification
- No dependency on centralized authority
- Transparent system

Steps to Connect MetaMask with Remix IDE:

1. Install MetaMask extension

2. Create a wallet account
3. Select Sepolia Testnet
4. Get free ETH from faucet
5. Open Remix IDE
6. Select “Injected Provider - MetaMask”
7. Connect MetaMask
8. Compile smart contract
9. Deploy contract
10. Confirm transaction
11. Interact with functions

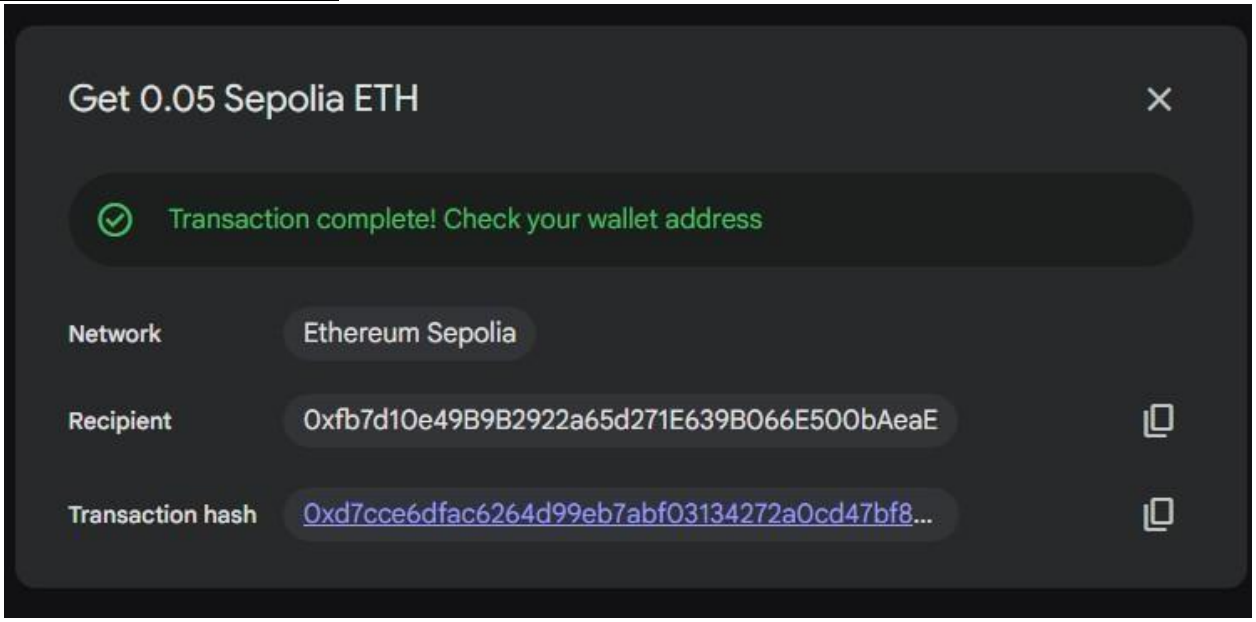
Program (Smart Contract):

```
// SPDX-License-Identifier: MIT pragma solidity
^0.8.0;

contract AcademicCredentials {

   struct Credential {
uint256 id;        string
studentName;        string
course;        string grade;
address issuedBy;
uint256 timestamp;
   }

   mapping(uint256 => Credential) public credentials;

   event CredentialAdded(
uint256 id,        string
studentName,        string
course,        string grade,
address issuedBy,        uint256
timestamp
   );

   function addCredential(        uint256
_id,        string memory
_studentName,        string memory
_course,
       string memory _grade
   ) public {

       require(credentials[_id].id == 0, "Already exists!");
```

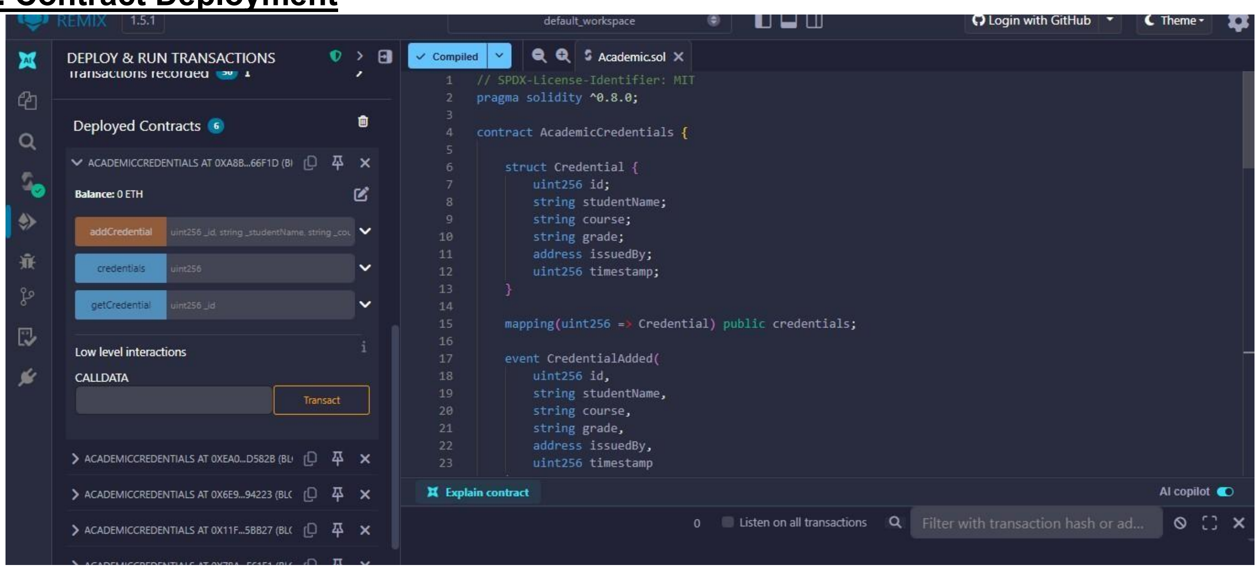
```
        credentials[_id] = Credential(
            _id,
            _studentName,
            _course,
_grade,             msg.sender,
block.timestamp
        );

        emit CredentialAdded(
            _id,
            _studentName,
            _course,
_grade,             msg.sender,
block.timestamp
        );
    }

    function getCredential(uint256 _id) public view returns (
uint256,         string memory,         string memory,         string
memory,         address,         uint256
    ) {
        Credential memory c = credentials[_id];         return
(             c.id,
            c.studentName,
            c.course,
            c.grade,
            c.issuedBy,
            c.timestamp
        );
    }
}
```

Output:

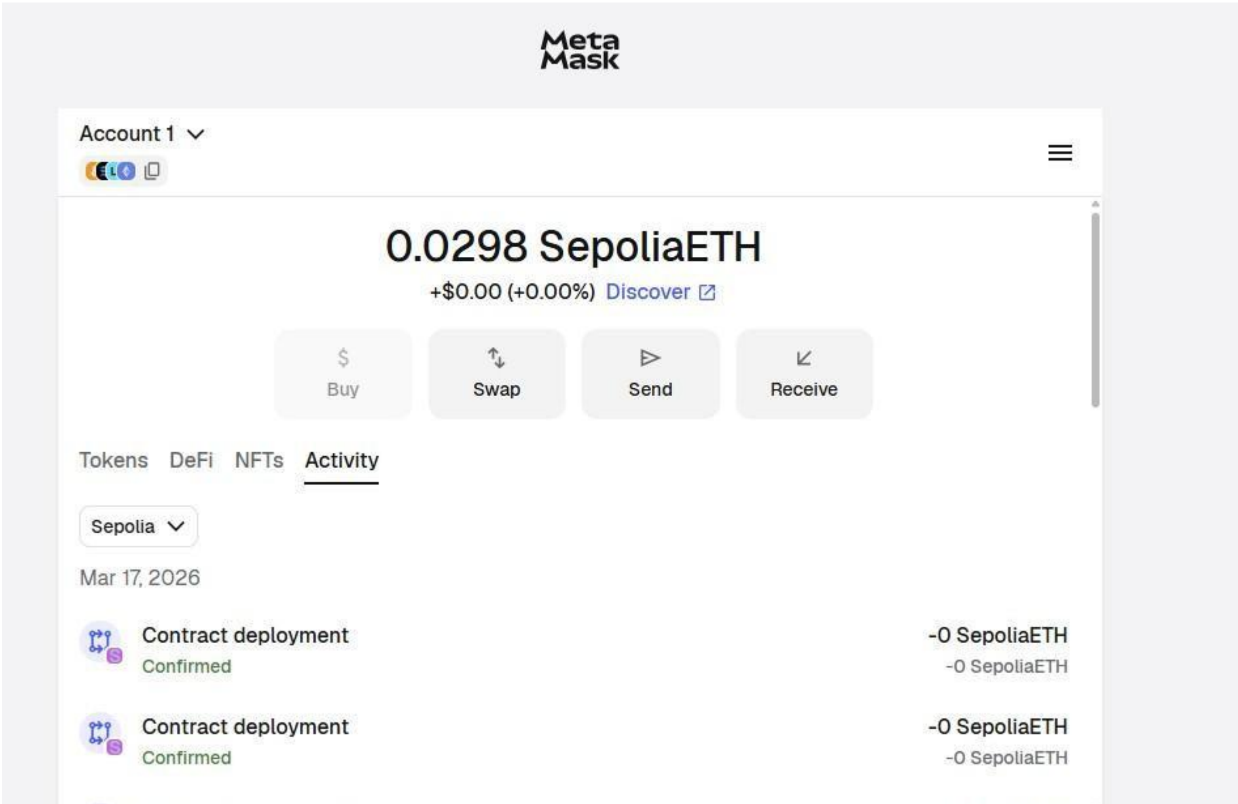
1. MetaMask Connected



2. Contract Deployment



3. Transaction Success



Conclusion:

The experiment successfully demonstrated the integration of MetaMask with Remix IDE and execution of blockchain transactions. The smart contract was deployed on Sepolia Testnet and used to store and verify academic credentials securely. This shows how blockchain can be used for tamper-proof academic verification systems.